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Background

Existing research has examined AI governance through various lenses; however, the relationship between economic freedom and AI-related policy discourse remains underexplored. This research seeks to address that gap by investigating whether a country's level of economic freedom shapes how governments frame the development and regulation of AI within national strategy documents.

In particular, this study compares the G7 countries and China to analyze how different economic and political environments influence AI regulatory narratives, policy priorities, and governance orientations.

? Research Question

Primary Research Question

How do variations in economic freedom across jurisdictions shape the dominant frames, justifications, and priorities embedded in national AI regulatory discourse for G7 countries, and how does this compare to China?

PROF

Alternative Research Question

To what extent do economic freedom and AI vibrancy predict the dominant frames embedded in national AI regulatory discourse, and how do these relationships vary across G7 countries and China?

Hypotheses

● H1 — AI Vibrancy and Innovation-Oriented Rhetoric

Countries with higher Stanford AI Vibrancy scores will exhibit more innovation-enabling rhetoric in their national policy documents.

Rationale

Countries with more developed AI ecosystems are expected to have stronger industry stakeholders and innovation-oriented policy agendas, encouraging governments to frame AI regulation in enabling rather than restrictive terms.

● H2 — Economic Freedom and Positive Regulatory Framing

Countries whose AI regulatory discourse is more positive and innovation-oriented will exhibit higher economic freedom scores.

Rationale

Positive regulatory framing may reflect broader ideological commitments toward market liberalism and economic openness, which are captured by economic freedom indicators.

● H3 — Regulatory Framing as a Mediating Mechanism

The relationship between AI vibrancy and economic freedom is partially or fully mediated by the permissiveness of AI regulatory rhetoric.

Rationale

Regulatory framing may function as an intermediate mechanism linking AI ecosystem maturity to broader economic ideology. In this view, AI vibrancy influences how governments discuss AI, while that discourse reflects and reinforces underlying economic values.

⚙ Methodology

1 Text Analysis

National AI strategy and regulatory documents from G7 countries, China, and the European Union will be collected and analyzed using Qwen sentence-level framing.

The analysis estimates the relative emphasis placed on AI-friendly versus AI-cautious discourse. AI-friendly framing includes language about innovation, adoption, investment, productivity, competitiveness, regulatory flexibility, and reduced regulatory burdens. AI-cautious framing includes language about risk, safety, oversight, compliance, accountability, privacy, fairness, rights protection, and precaution.

China is treated as one country-level case and combines both the national AI strategy document and the newer State Council “AI+” action policy. Each country/entity receives confidence-weighted framing scores normalized per 1,000 words.

For consistency, framing analysis uses an English analysis corpus. Chinese-heavy text is translated into English with [facebook/nllb-200-distilled-600M](#); English source documents are used directly.

Correlation Analysis

Correlation analysis is conducted in [notebooks/Correlation Study/correlation_study.ipynb](#) to examine relationships between:

-  Economic Freedom scores and regulatory framing scores
-  Stanford AI Vibrancy scores and regulatory framing scores
-  Combined AI Vibrancy + Economic Freedom predictors and the innovation-to-risk framing ratio

The core dependent variable is the **Innovation-to-Risk Framing Ratio**, calculated as the ratio of Qwen-classified AI-friendly framing to Qwen-classified AI-cautious framing. The notebook uses the Qwen framing output by default and stores a merged cross-country dataset, correlation table, regression models, mediation diagnostic, and visual summaries in [outputs/](#).

Expected Contribution

This research contributes to the emerging literature on AI governance by connecting political-economic structure with regulatory discourse.

Rather than treating AI policy solely as a technical or legal issue, this study examines how broader economic ideology may shape national narratives surrounding AI development and regulation.

Core Analytical Logic

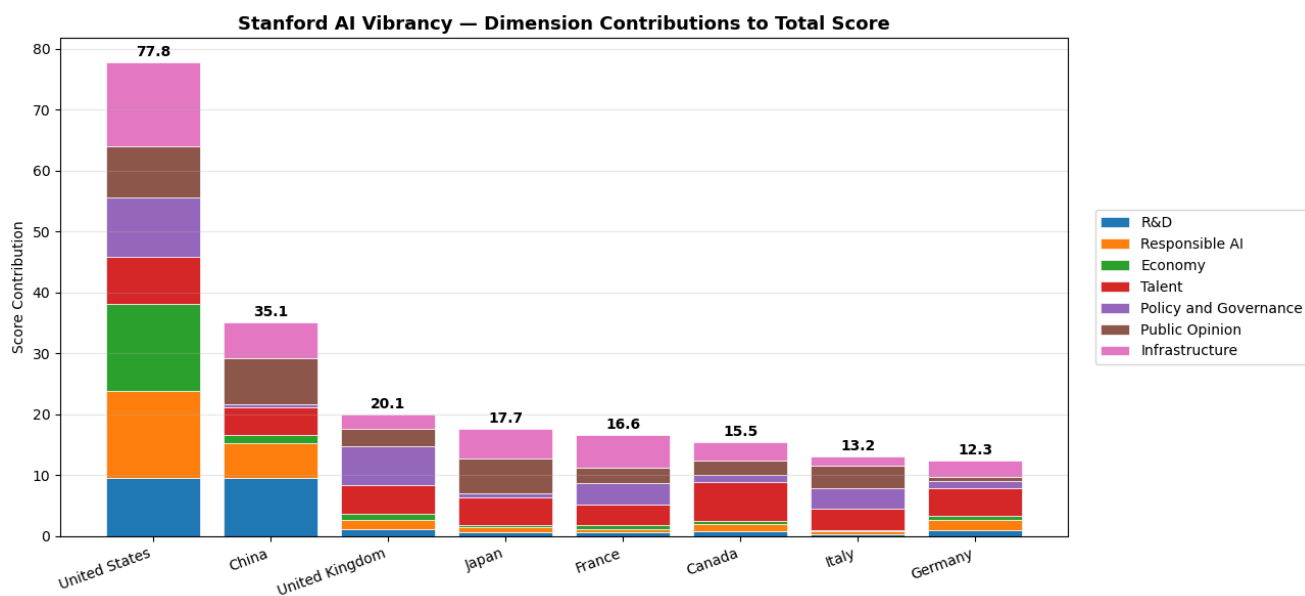
```
AI Vibrancy
  ↓
Regulatory Framing
  ↓
Economic Freedom Orientation
```

This framework explores whether AI ecosystem maturity influences regulatory rhetoric, and whether that rhetoric reflects broader economic and ideological preferences.

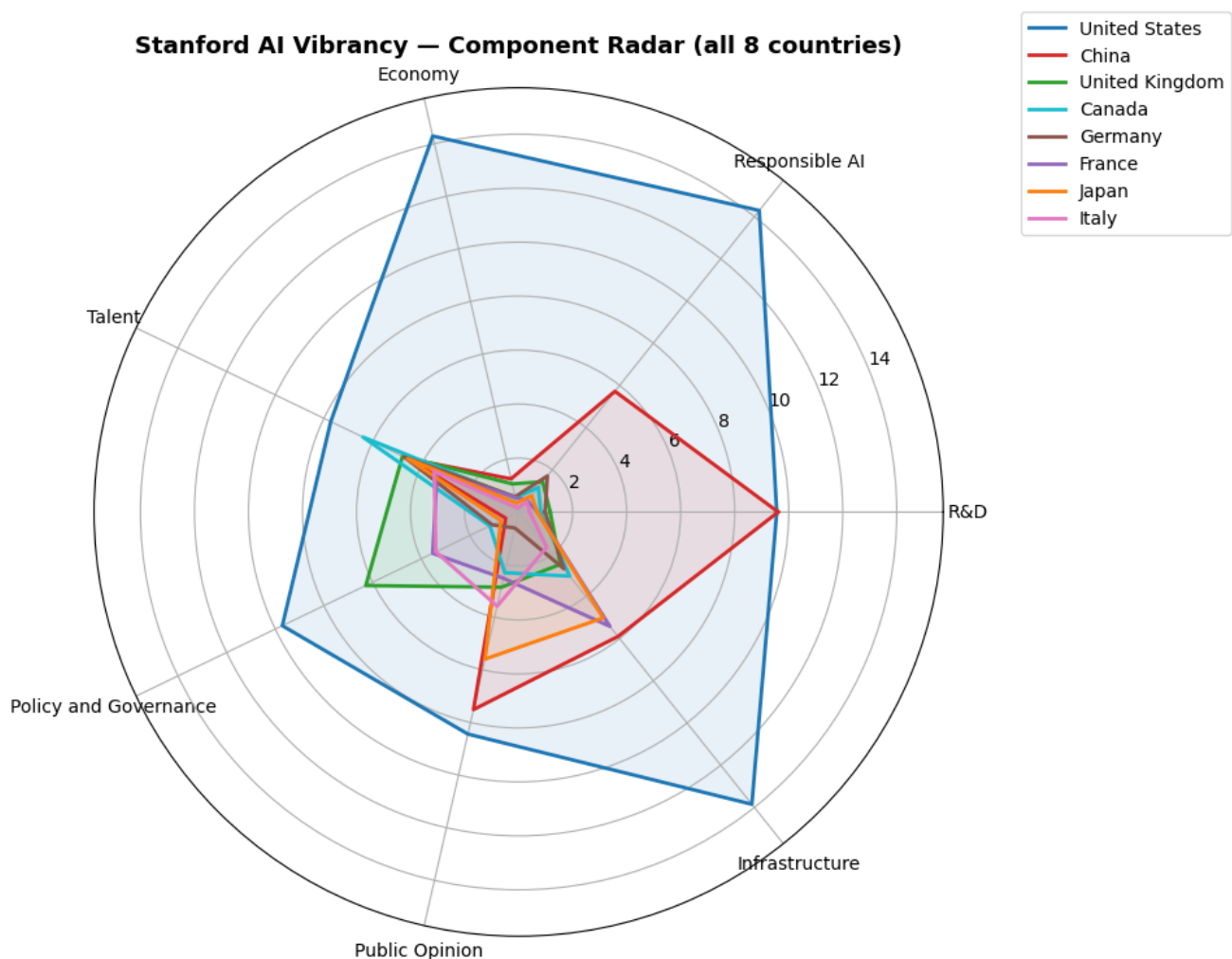
Preliminary Results

Stanford AI Vibrancy Tool — G7 + China

Cross-country comparison of AI vibrancy scores (Total Score and 7 sub-dimensions: R&D, Responsible AI, Economy, Talent, Policy and Governance, Public Opinion, Infrastructure).



Stanford AI Vibrancy — Component Radar (all 8 countries)



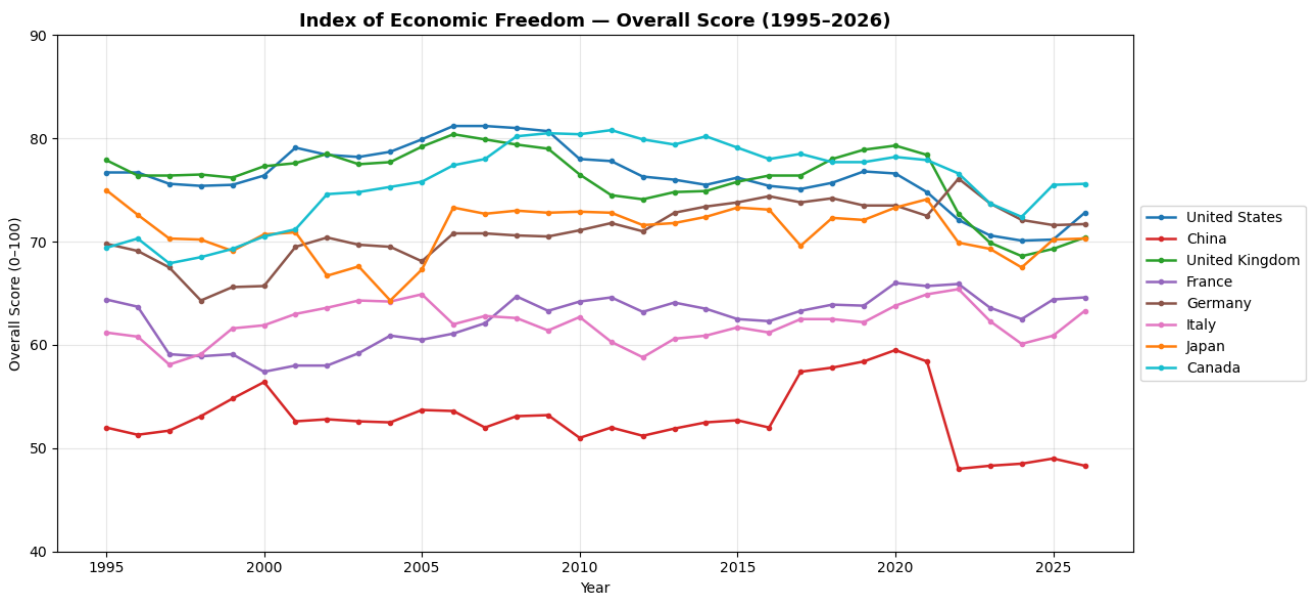
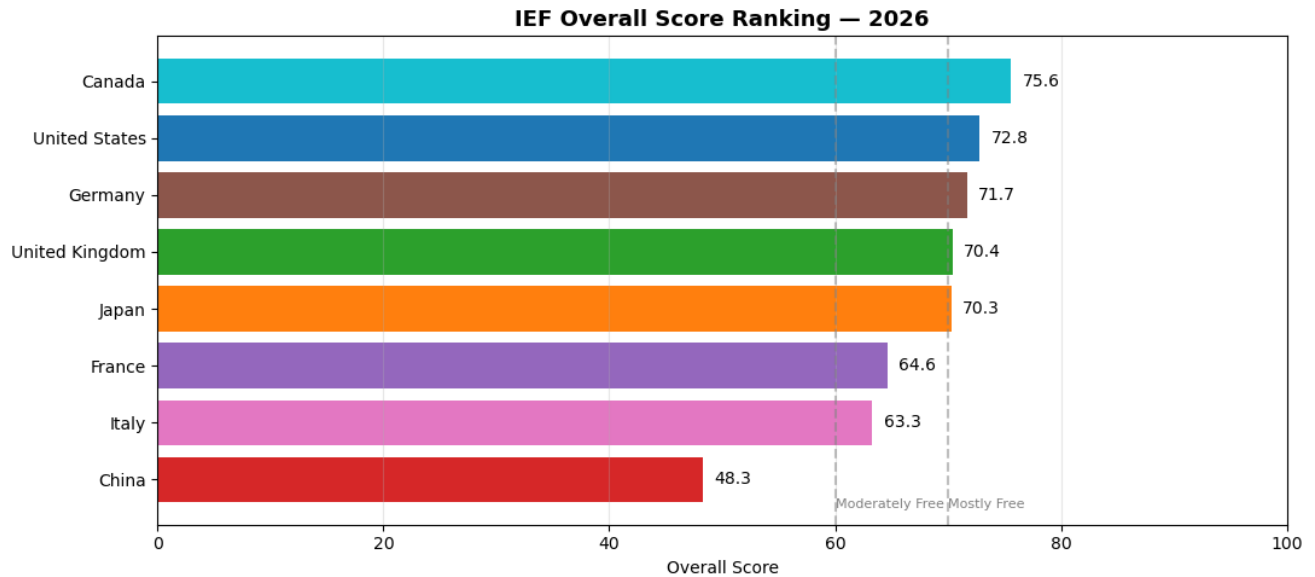
Key observations:

- The United States leads overall (Total Score ≈ 77.85), driven primarily by Responsible AI, Economy, and Infrastructure.
- China ranks second (≈ 35.10) — its R&D score is essentially tied with the U.S., but it lags substantially in Responsible AI, Economy, and Policy & Governance.
- The remaining G7 countries cluster between 12 and 21, with no single G7 member dominating.

- The component radar chart highlights how each country's AI vibrancy profile differs across R&D, Responsible AI, Economy, Talent, Policy and Governance, Public Opinion, and Infrastructure.

Index of Economic Freedom — G7 + China

Heritage Foundation Economic Freedom scores for the same eight countries, used as the political-economic context for the framing analysis.



Key observations:

- G7 economies generally cluster in the "Mostly Free" / "Moderately Free" range.
- China sits noticeably below the G7, providing a natural contrast for testing whether economic freedom levels relate to AI regulatory framing.
- The yearly trend chart shows how economic freedom scores change over time, providing temporal context for the cross-sectional comparison.

Descriptive Statistics — Key Numeric Variables

To address the feedback, the project now explicitly reports standard descriptive statistics for the main numeric variables. The tables below report count, mean, median, standard deviation, minimum, and maximum for the Heritage Index of Economic Freedom scores and Stanford AI Vibrancy metrics.

Index of Economic Freedom

Variable	count	mean	median	stdev	min	max
Overall Score	256	68.48	70.3	8.49	48.0	81.2
Property Rights	256	77.09	85.95	20.04	20.0	96.2
Government Integrity	256	70.47	76.0	17.9	22.0	95.1
Judicial Effectiveness	80	76.22	76.8	14.36	37.4	97.9
Tax Burden	256	61.41	62.35	10.62	33.2	80.0
Government Spending	256	43.14	43.95	22.77	0.0	95.9
Fiscal Health	80	48.77	54.55	30.69	0.0	92.9
Business Freedom	256	79.47	83.5	11.59	43.1	100.0
Labor Freedom	176	67.27	68.05	14.97	39.9	98.5
Monetary Freedom	256	81.03	81.5	5.96	61.5	94.3
Trade Freedom	256	79.42	80.8	9.94	20.0	88.8
Investment Freedom	256	66.19	70.0	18.3	20.0	90.0
Financial Freedom	256	63.12	70.0	17.72	20.0	90.0

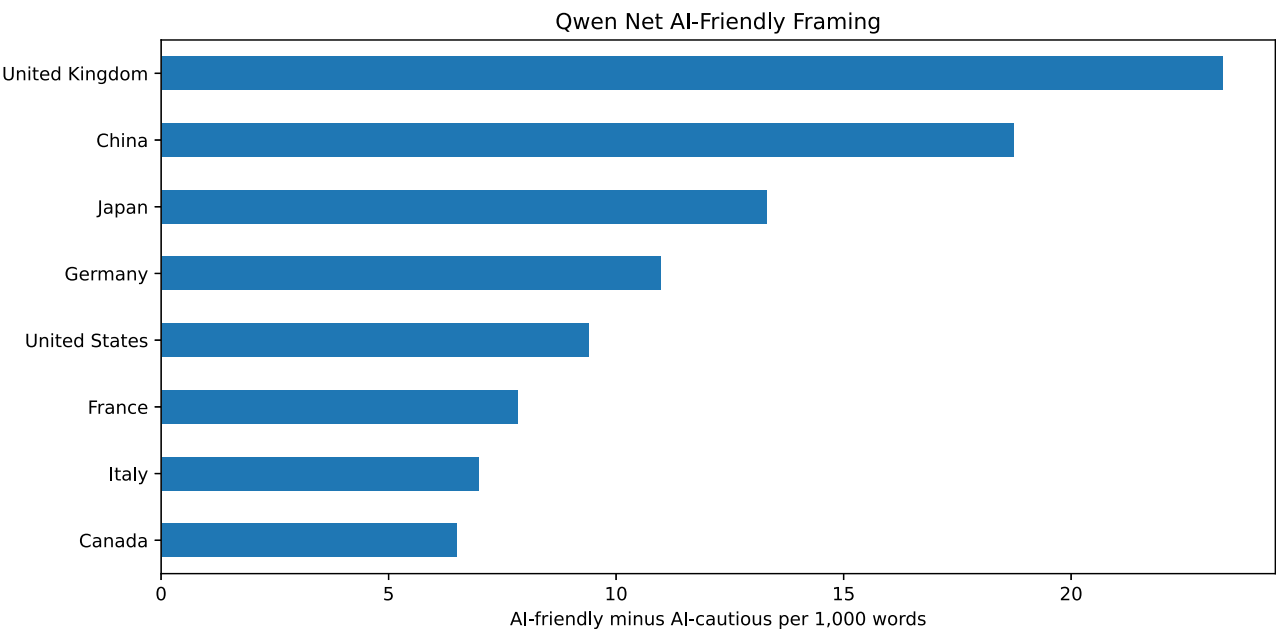
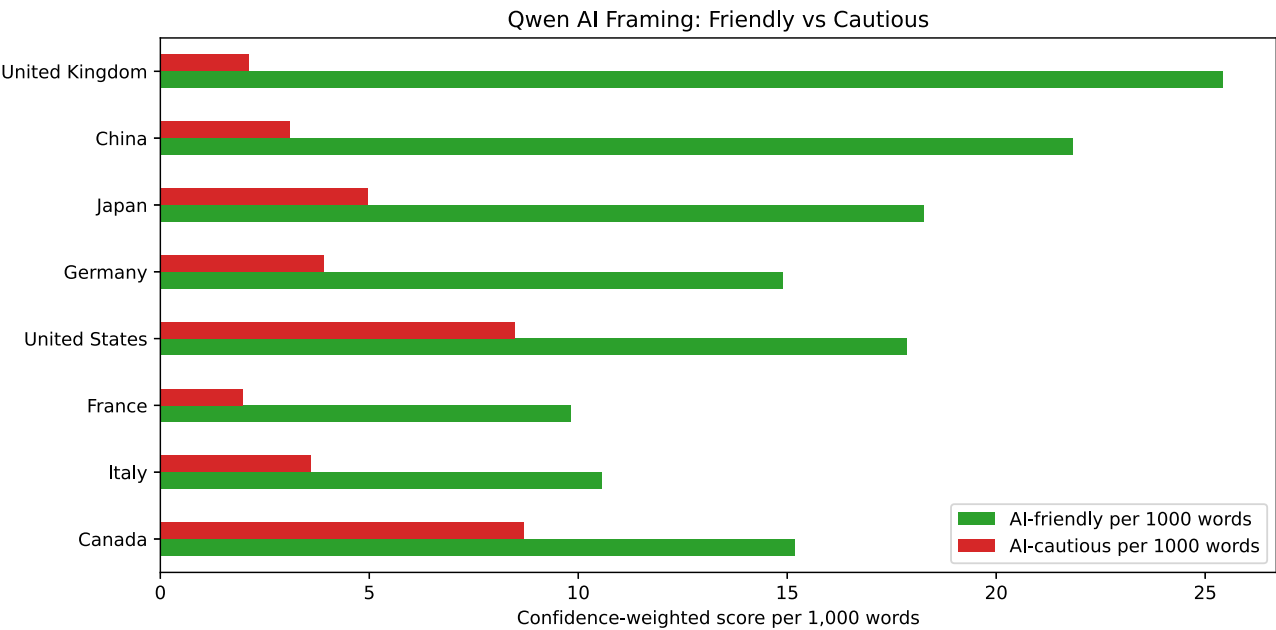
Stanford AI Vibrancy

Variable	count	mean	median	stdev	min	max
Total Score	8	26.04	17.16	22.12	12.35	77.85
R&D	8	2.99	0.9	4.08	0.41	9.61
Responsible AI	8	3.26	1.3	4.77	0.49	14.29
Economy	8	2.35	0.57	4.84	0.15	14.29
Talent	8	4.94	4.63	1.46	3.35	7.72
Policy and Governance	8	3.31	2.29	3.24	0.53	9.71
Public Opinion	8	4.18	3.22	2.73	0.61	8.44
Infrastructure	8	5.01	4.03	3.89	1.67	13.85

The CSV versions are saved in [outputs/descriptive_statistics_ief.csv](#), [outputs/descriptive_statistics_ai_vibrancy.csv](#), and [outputs/descriptive_statistics_key_numeric_variables.csv](#).

Qwen AI-Friendly vs. AI-Cautious Framing

The Qwen sentence-level classifier labels each policy sentence as AI-friendly, AI-cautious, mixed, or neutral. AI-friendly sentences emphasize innovation, adoption, investment, productivity, competitiveness, regulatory flexibility, or faster deployment. AI-cautious sentences emphasize risk, safety, oversight, compliance, accountability, privacy, fairness, rights protection, or precaution. Mixed sentences contribute half of their confidence score to each side, and neutral sentences contribute zero.



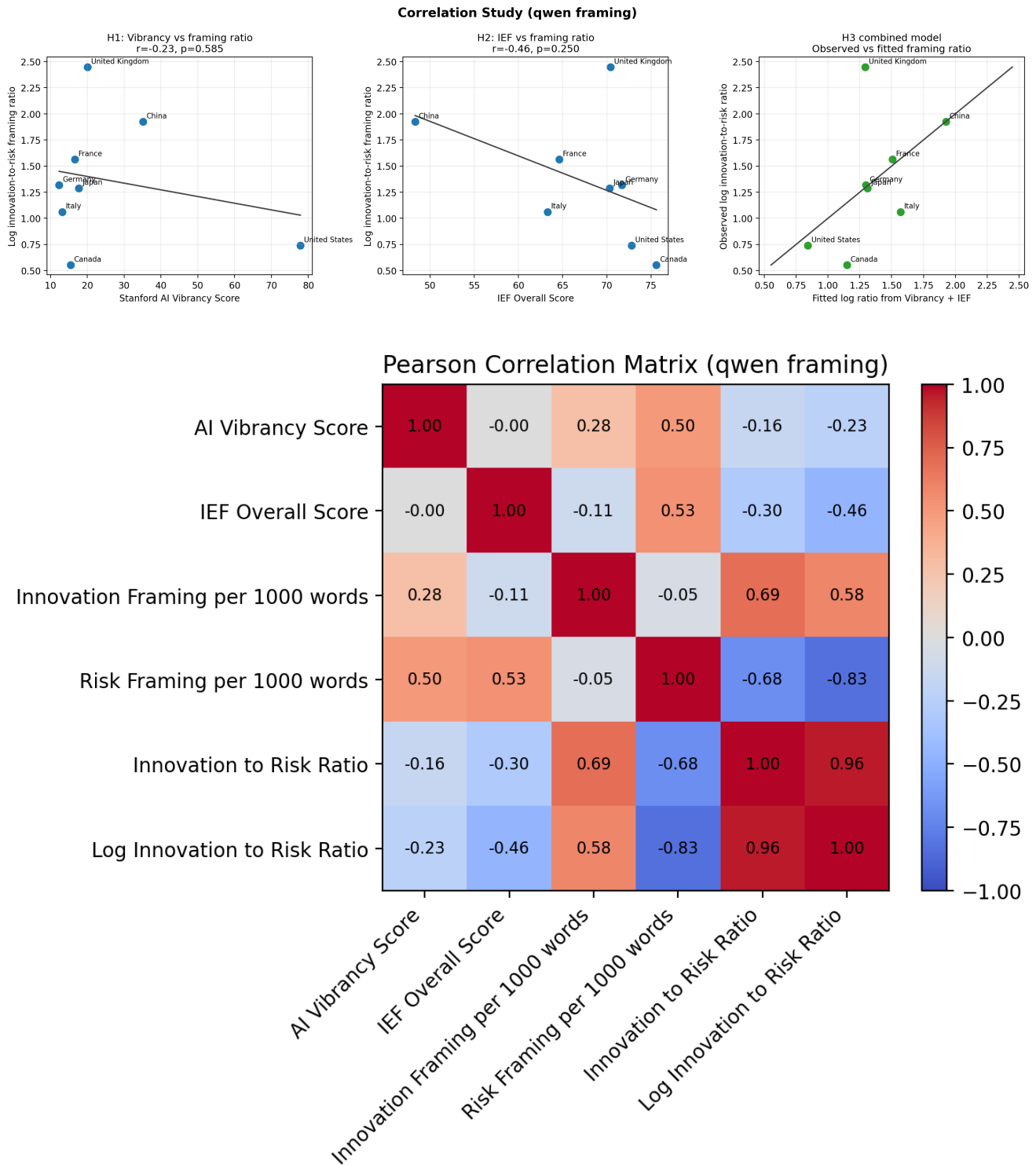
Outputs are saved in [outputs/qwen_ai_framing_summary.csv](#), [outputs/qwen_ai_framing_sentence_labels.csv](#), [outputs/qwen_g7_china_framing_table.csv](#), and [outputs/qwen_ai_framing_overview.md](#).

These visualizations together provide the empirical anchor for the hypotheses (H1–H3): the variation in AI vibrancy, economic freedom, and regulatory framing across the analyzed jurisdictions is large enough

to test whether broader political-economic and AI ecosystem conditions predict the framing of national AI strategies.

Correlation Study — Vibrancy, Economic Freedom, and Framing

The correlation study combines Stanford AI Vibrancy scores, 2026 Heritage Index of Economic Freedom scores, and the Qwen framing results for the same G7 + China country set. The main framing variable is the log innovation-to-risk ratio, derived from Qwen AI-friendly framing per 1,000 words divided by Qwen AI-cautious framing per 1,000 words.



Initial exploratory results:

- H1 shows a weak negative relationship between AI Vibrancy and the Qwen log innovation-to-risk framing ratio (Pearson $r = -0.229$, $p = 0.5853$).
- H2 shows a negative relationship between IEF score and the Qwen log innovation-to-risk framing ratio (Pearson $r = -0.461$, $p = 0.2504$), so the current G7 + China sample does not support the expected positive direction.
- The combined model using AI Vibrancy and IEF explains more variation than AI Vibrancy alone ($R^2 = 0.265$ versus $R^2 = 0.052$), but with only eight countries the results should be treated as exploratory rather than confirmatory.
- The mediation diagnostic is included as a mechanism check, not a formal causal test, because the sample size is too small for strong inference.

Key outputs are saved in `outputs/correlation_study_dataset_qwen.csv`, `outputs/correlation_study_correlations_qwen.csv`, `outputs/correlation_study_regression_models_qwen.csv`, and `outputs/correlation_study_mediation_qwen.csv`.

Folder Structure

```

AI-Policy/
├── data/
│   ├── AI vibrancy tool screen shot/
│   │   ├── ai_country_scores.csv
│   │   ├── ai_country_scores_long.csv
│   │   └── *.png # Stanford AI Vibrancy source
├── screenshots
│   ├── Index of Economic Freedom/
│   │   ├── G7+China All Data
│   │   └── country-level IEF files
│   ├── number/
│   │   ├── ai_dev_index_g7_china.csv
│   │   ├── g7_china_datasets.csv
│   │   ├── ief_g7_china_panel.csv
│   │   └── PUBLIC DATA_ 2026 AI INDEX REPORT/
│   └── pdf/
│       ├── AI Policy/
│       │   ├── CA_National_AI_Strategy.pdf
│       │   ├── CA_National_AI_Strategy_ocr.pdf
│       │   ├── CN_National_AI_Strategy.pdf
│       │   ├── DE_National_AI_Strategy.pdf
│       │   ├── EU_AI_Strategy.pdf
│       │   ├── FR_National_AI_Strategy.pdf
│       │   ├── IT_National_AI_Strategy.pdf
│       │   ├── JP_National_AI_Strategy.pdf
│       │   ├── JP_National_AI_Strategy_ocr.pdf
│       │   ├── UK_National_AI_Strategy.pdf
│       │   ├── UK_National_AI_Strategy_ocr.pdf
│       │   ├── US_National_AI_Strategy.pdf
│       │   ├── 国务院关于深入实施“人工智能+”行动的意见.pdf
│       │   └── _extracted/

```

- ├── *.txt # Extracted strategy text by country/entity
- ├── document_sources.csv
- ├── document_stats.csv
- ├── theme_counts_raw.csv
- ├── theme_per_1000_words.csv
- ├── framing_keyword_counts.csv
- ├── framing_category_counts.csv
- ├── framing_scores_per_1000_words.csv
- ├── framing_share_of_mentions.csv
- ├── keyword_sentence_matches.csv
- ├── similarity_matrix.csv
- └── _translated_nllb/
 - ├── China.txt # NLLB English translation for Chinese-heavy policy text
 - └── translation_stats.csv
- ├── notebooks/
 - ├── AI Development/
 - ├── ai_vibrancy_tool.ipynb
 - ├── ai_dev_index_v2.ipynb
 - └── select_g7_china_data.ipynb
 - ├── Correlation Study/
 - └── correlation_study.ipynb
 - ├── Economic Freedom/
 - └── ief_g7_china_analysis.ipynb
 - └── Text Analysis/
 - ├── national_ai_strategy_nllb_translation.ipynb
 - ├── national_ai_strategy_text_analysis.ipynb
 - └── qwen_ai_framing.ipynb
- ├── outputs/
 - ├── stanford_AI_Vibrancy.png
 - ├── stanford_AI_Vibrancy_component_radar.png
 - ├── IEF_score.png
 - ├── IEF_score_by_year.png
 - ├── qwen_ai_framing_summary.csv
 - ├── qwen_ai_framing_sentence_labels.csv
 - ├── correlation_study_dataset_qwen.csv
 - ├── correlation_study_correlations_qwen.csv
 - ├── correlation_study_regression_models_qwen.csv
 - ├── correlation_study_mediation_qwen.csv
 - ├── correlation_study_scatter_qwen.png
 - └── correlation_study_matrix_qwen.png
- ├── README.md
- ├── README.pdf
- └── LICENSE